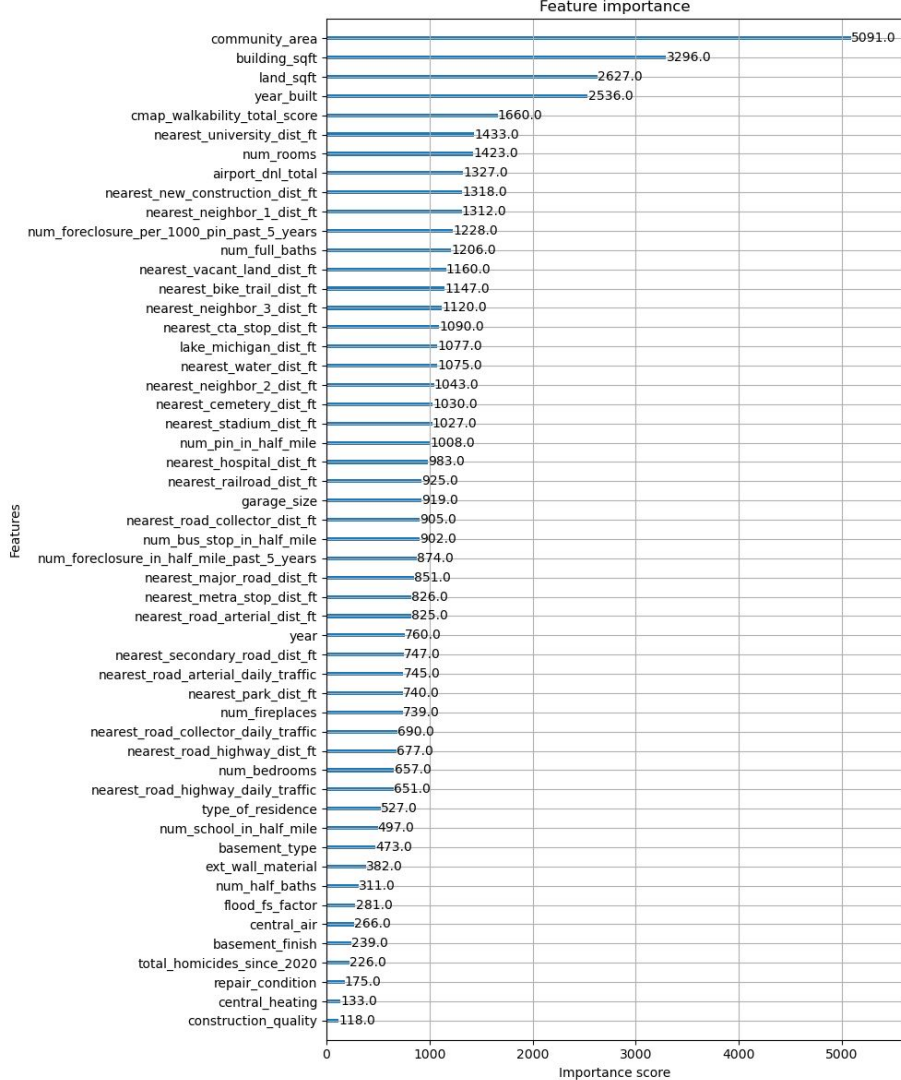


Chicago Neighborhood Premiums

Nicholas Birmingham, Vijay Dabhi,
Noah Riggensbach, and Bradley Zykoski

Community Areas in Chicago

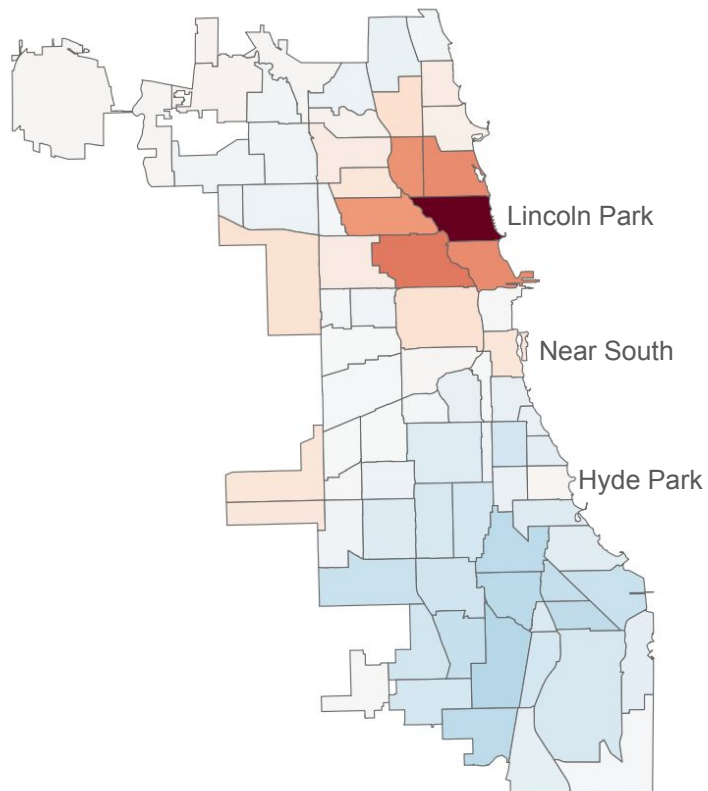
- Chicago is made up of 77 “community areas” (i.e. neighborhoods)
- Community areas are quite different from each other, even those that are directly adjacent: This affects housing prices
- We determined the relative premium of each community area for single-family homes



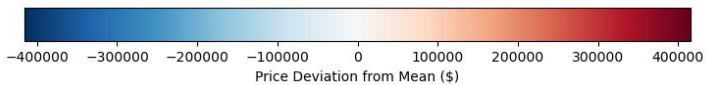
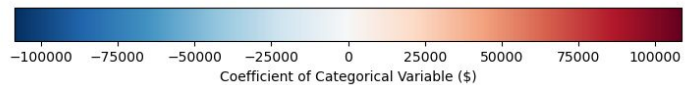
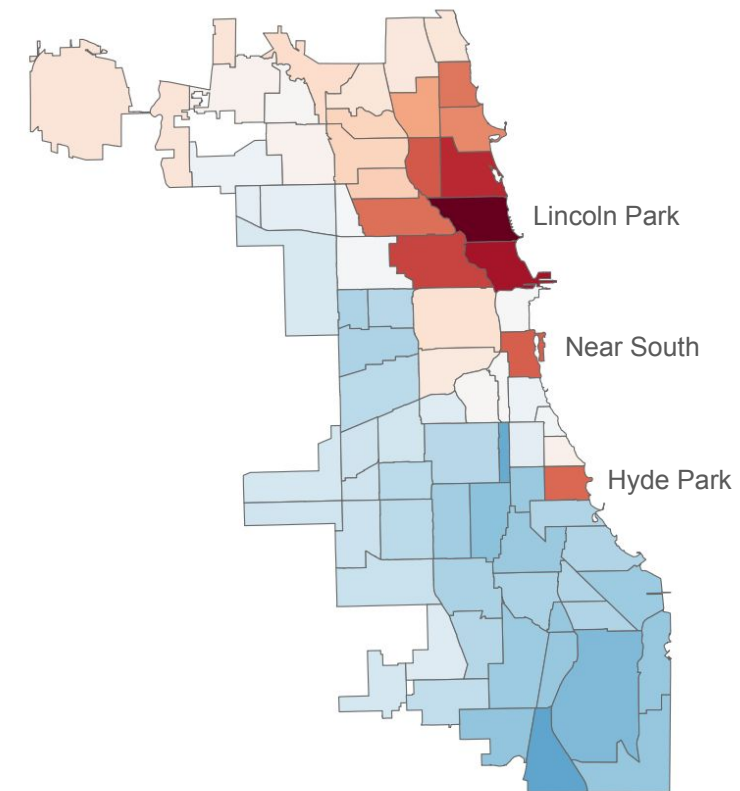
Our Data

- Property sales data from Cook County Assessor's Office
- Homicide data from Chicago Police Department

**Linear Model Partial Dependence:
Significance of Community Area in Housing Price**

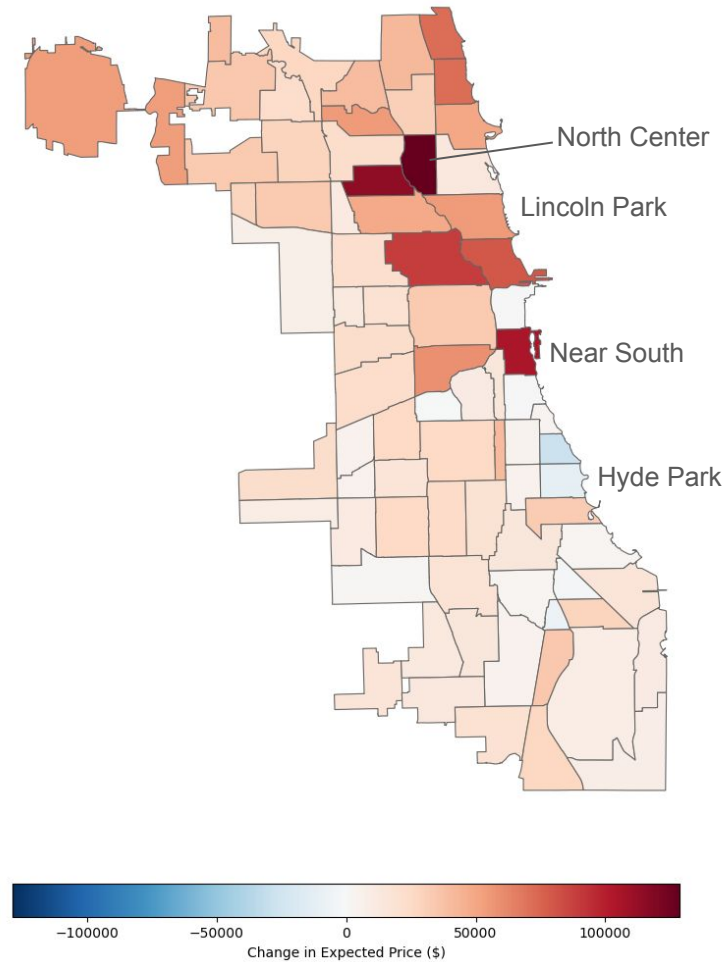


**Gradient Boosted Tree Partial Dependence:
Significance of Community Area in Housing Price**



**Shift in Neighborhood Price Impact
(2020-2023 vs 2023-Present)**

Gentrification



Second business application: North Center Gentrification

- Two investors want to buy a property in the north side of Chicago in 2024 and sell in 2025
- First buys in Lincoln Park, second buys in North Center based on our analysis.
 - The first investor makes an average of \$235,011.54
 - The second investor makes an average of \$281,616.61
- The second investor who used our model makes on average \$46,605.07 more.

Future Directions

- Update model to account for distinct lowest-decile sales trends
- Collect demographic data from census